

# Deep Generative Models

## Lecture 10

Roman Isachenko



AI Masters

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# Recap of Previous Lecture

## DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[ \frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t} \left\| \mathbf{s}_{\theta, t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]$$

In practice, **this coefficient** is usually omitted.

## NCSN Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

**Note:** The objectives of DDPM and NCSN are almost identical; however, their sampling procedures differ:

- ▶ NCSN utilizes annealed Langevin dynamics,
- ▶ DDPM employs ancestral sampling.

# Recap of Previous Lecture

## Unconditional Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \sigma_t \cdot \epsilon$$

## Conditional Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) + \sigma_t \cdot \epsilon$$

## Conditional Distribution

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \\ &= \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)\end{aligned}$$

Here,  $p(\mathbf{y}|\mathbf{x}_t)$  denotes a classifier operating on noisy samples (which must be trained separately).

# Recap of Previous Lecture

## Guided Score Function

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

**Note:** Increasing  $\gamma$  sharpens  $p(\mathbf{y}|\mathbf{x}_t)$ , increasing contrast

$$\hat{p}(\mathbf{y}|\mathbf{x}_t) \propto p(\mathbf{y}|\mathbf{x}_t)^{\gamma}.$$

## Training

- ▶ Train the DDPM as before.
- ▶ Train an additional classifier  $p(\mathbf{y}|\mathbf{x}_t)$  on noisy data.

## Guided Sampling

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon$$

## Recap of Previous Lecture

The previous method requires an additional classifier  $p(\mathbf{y}|\mathbf{x}_t)$  trained on noisy data. Let's try to avoid this requirement.

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t)$$

$$\begin{aligned}\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) = \\ &= (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y})\end{aligned}$$

## Scaled Guided Score Function

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

- ▶ Introduce  $\mathbf{y} = \emptyset$  for  $\mathbf{s}_{\theta,t}(\mathbf{x}_t) = \mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$ .
- ▶ Drop the labels  $\mathbf{y}$  with some fixed probability.
- ▶ Apply the model twice during inference to get  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$  and  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ .

# Recap of Previous Lecture

## Continuous-Time Dynamics (Neural ODE)

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t); \quad \text{where } \mathbf{x}(t_0) = \mathbf{x}_0.$$

$$\psi_t(\mathbf{x}_0) = \int_{t_0}^{t_1} \mathbf{v}_{\theta}(\mathbf{x}(t), t) dt + \mathbf{x}_0 \approx \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0, t_1).$$

- ▶  $\mathbf{v}_{\theta} : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$  is a vector field.
- ▶  $\psi : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$  is a flow (the solution of ODE):

## Euler Update Step (ODESolve)

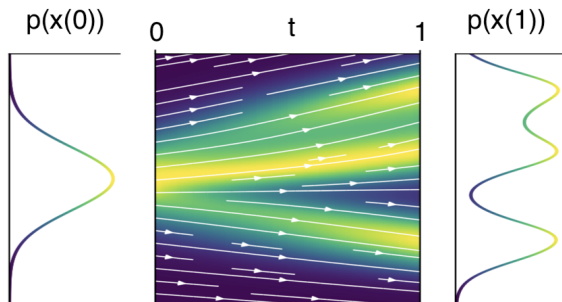
$$\mathbf{x}(t+h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

More advanced numerical methods (such as Runge-Kutta) are often used instead of unstable Euler update step.

# Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_\theta(\mathbf{x}(t), t); \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

- ▶ Suppose  $\mathbf{x}(0)$  is a random variable with density  $p_0(\mathbf{x})$ . Then,  $\mathbf{x}(t)$  is a random variable with density  $p_t(\mathbf{x})$ .
- ▶  $p_t(\mathbf{x}) = p(\mathbf{x}, t)$  describes the **probability path** between  $p_0(\mathbf{x})$  and  $p_1(\mathbf{x})$ .



# Recap of Previous Lecture

## Theorem (Picard)

If  $\mathbf{v}$  is uniformly Lipschitz continuous in  $\mathbf{x}$  and continuous in  $t$ , then the ODE has a **unique** solution.

This means the ODE can be **uniquely inverted**:

$$\begin{aligned}\mathbf{x}(1) &= \mathbf{x}(0) + \int_0^1 \mathbf{v}_\theta(\mathbf{x}(t), t) dt \\ \mathbf{x}(0) &= \mathbf{x}(1) + \int_1^0 \mathbf{v}_\theta(\mathbf{x}(t), t) dt\end{aligned}$$

**Note:** Unlike discrete-time NF,  $\mathbf{v}$  need not be invertible (uniqueness ensures bijectivity).

How can we compute  $p_t(\mathbf{x})$  for any  $t$ ?

# Outline

1. Continuity Equation for NF Log-Likelihood
2. SDE Basics
3. Diffusion and Score Matching SDEs
4. Probability Flow ODE

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# Continuous-Time NF

## Theorem (Continuity Equation)

If  $\mathbf{v}$  is uniformly Lipschitz continuous in  $\mathbf{x}$  and continuous in  $t$ , then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

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This result states: given  $\mathbf{x}_0 = \mathbf{x}(0)$ , the solution to the continuity equation gives the density  $p_1(\mathbf{x}(1))$ .

## Solution of the Continuity Equation

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \text{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt.$$

# Continuous-Time NF

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- ▶ This provides the density along the trajectory (not the total probability path).
- ▶ However, **the latter term** is difficult to estimate efficiently.

# Outline

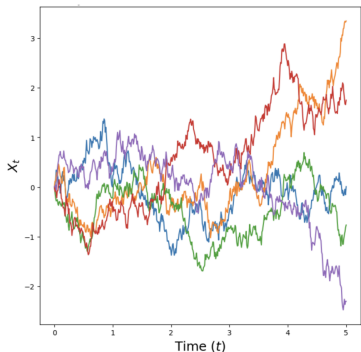
1. Continuity Equation for NF Log-Likelihood
2. SDE Basics
3. Diffusion and Score Matching SDEs
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# Stochastic Differential Equation (SDE)

## Wiener Process

$\mathbf{w}(t)$  is the standard Wiener process (Brownian motion), defined by:

1.  $\mathbf{w}(0) = 0$  (almost surely);
2.  $\mathbf{w}(t)$  has independent increments;
3.  $\mathbf{w}(t)$  trajectories are continuous;
4.  $\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t-s)\mathbf{I})$  for  $t > s$ ;

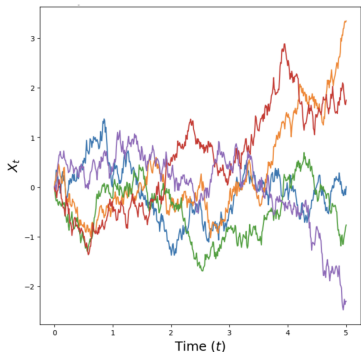


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$d\mathbf{w} = \mathbf{w}(t + dt) - \mathbf{w}(t) = \mathcal{N}(0, \mathbf{I} \cdot dt) = \epsilon \cdot \sqrt{dt}$ , where  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .

# Stochastic Differential Equation (SDE)

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Let's define a stochastic process  $\mathbf{x}(t)$  with initial condition  $\mathbf{x}(0) \sim p_0(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$ :

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

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- ▶  $\mathbf{f}(\mathbf{x}, t) : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}^m$  is the **drift** term (vector field).
- ▶  $g(t) : \mathbb{R} \rightarrow \mathbb{R}$  is the **diffusion** term (if  $g(t) = 0$ , we recover the standard ODE).
- ▶  $\mathbf{w}(t)$  is the standard Wiener process ( $d\mathbf{w} = \epsilon \cdot \sqrt{dt}$ ).
- ▶ We do not have the flow  $\psi_t(\mathbf{x}_0)$  notion anymore, since trajectories are stochastic.

# Itô Integration Basics

The SDE should be understood in its **integral form** (Itô sense):

$$\mathbf{x}(t) = \mathbf{x}(0) + \int_0^t \mathbf{f}(\mathbf{x}(s), s) ds + \int_0^t g(s) d\mathbf{w}(s)$$

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- ▶ The first integral is a classical (Riemann) integral.
- ▶ The second integral is an **Itô stochastic integral**.

# Itô Integration Basics

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- ▶ The **first integral** is a classical (Riemann) integral.
- ▶ The **second integral** is an **Itô stochastic integral**.

$$\int_0^t g(s) d\mathbf{w}(s) = \lim_{N \rightarrow \infty} \sum_{i=0}^{N-1} g(t_i) (\mathbf{w}(t_{i+1}) - \mathbf{w}(t_i))$$

- ▶ Brownian paths are continuous but a.s. **nowhere differentiable**  $\Rightarrow$  Fundamental Theorem of Calculus fails.
- ▶ Instead, SDEs require **Itô calculus** (e.g., Itô's lemma).
- ▶ Expressions  $d\mathbf{x}$ ,  $dt$ ,  $d\mathbf{w}$  are **formal shorthand** for infinitesimal increments.

# Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

## Theorem

If  $\mathbf{f}$  is continuously differentiable with a bounded derivative in  $\mathbf{x}$  and continuous in  $t$  and  $g(t)$  is continuous then the SDE has the solution given by unique process  $\mathbf{x}(t)$ .

- ▶ Unlike ODEs, the initial condition  $\mathbf{x}(0)$  doesn't uniquely determine the trajectory.
- ▶ There are two sources of randomness:
  - ▶ the initial distribution  $p_0(\mathbf{x})$ ;
  - ▶ the Wiener process  $\mathbf{w}(t)$ .

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Discretizing the SDE (Euler-Maruyama Update) – SDEsolve

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \epsilon \cdot \sqrt{dt}$$

If  $dt = 1$ , then

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{f}(\mathbf{x}_t, t) + g(t) \cdot \epsilon$$

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- ▶ At any time  $t$ , the process has density  $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ .
- ▶  $p : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}_+$  specifies a **probability path** from  $p_0(\mathbf{x})$  to  $p_1(\mathbf{x})$ .
- ▶ How can we obtain the probability path  $p_t(\mathbf{x})$  for  $\mathbf{x}(t)$ ?

# Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}, \quad d\mathbf{w} = \boldsymbol{\epsilon} \cdot \sqrt{dt}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

## Theorem (Kolmogorov-Fokker-Planck)

The evolution of  $p_t(\mathbf{x})$  is governed by

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

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Here,

$$\operatorname{div}(\mathbf{v}) = \sum_{i=1}^m \frac{\partial v_i(\mathbf{x})}{\partial x_i} = \operatorname{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x})}{\partial \mathbf{x}} \right)$$

$$\Delta_{\mathbf{x}}p_t(\mathbf{x}) = \sum_{i=1}^m \frac{\partial^2 p_t(\mathbf{x})}{\partial x_i^2} = \operatorname{tr} \left( \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right)$$

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- ▶ The KFP theorem is a necessary and sufficient condition (it uniquely defines  $p_t(\mathbf{x})$ ).
- ▶ This generalizes the continuity equation for continuous-time NF:

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x}, t)}{\partial \mathbf{x}} \right).$$

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## Special Case: constant density $p_t(\mathbf{x})$

Let find the SDE for which  $p_t(\mathbf{x}) = \text{const}$  (i.e., if  $\mathbf{x}(0) \sim p_0(\mathbf{x})$ , then  $\mathbf{x}(t) \sim p_0(\mathbf{x})$ ).

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$$\frac{\partial p_t(\mathbf{x})}{\partial t} = 0 \quad \Leftrightarrow \quad \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})] + \frac{1}{2} g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = 0$$

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$$\mathbf{f}(\mathbf{x}, t) = \frac{1}{2} g^2(t) \frac{1}{p_t(\mathbf{x})} \frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} = \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$

## Langevin SDE (Special Case)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = 0 \quad \Leftrightarrow \quad \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})] + \frac{1}{2} g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = 0$$

$$\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})] = \frac{1}{2} g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}$$

$$\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x}) = \frac{1}{2} g^2(t) \frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}}$$

$$\mathbf{f}(\mathbf{x}, t) = \frac{1}{2} g^2(t) \frac{1}{p_t(\mathbf{x})} \frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} = \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$

Let  $g(t) = 1$ , then  $\mathbf{f}(\mathbf{x}, t) = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$ .

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) dt + \mathbf{1} \cdot d\mathbf{w}$$

## Langevin SDE (Special Case)

Let find the SDE for which  $p_t(\mathbf{x}) = \text{const}$  (i.e., if  $\mathbf{x}(0) \sim p_0(\mathbf{x})$ , then  $\mathbf{x}(t) \sim p_0(\mathbf{x})$ ).

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) dt + \mathbf{1} \cdot d\mathbf{w}$$

## Langevin SDE (Special Case)

Let find the SDE for which  $p_t(\mathbf{x}) = \text{const}$  (i.e., if  $\mathbf{x}(0) \sim p_0(\mathbf{x})$ , then  $\mathbf{x}(t) \sim p_0(\mathbf{x})$ ).

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) dt + \mathbf{1} \cdot d\mathbf{w}$$

### Discretized Langevin SDE

$$\mathbf{x}_{t+1} - \mathbf{x}_t = \frac{\eta}{2} \cdot \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) + \sqrt{\eta} \cdot \epsilon, \quad \eta \approx dt.$$

### Langevin Dynamic

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) + \sqrt{\eta} \cdot \epsilon, \quad \eta \approx dt.$$

We (partially) explained, why Langevin dynamics is working.

# Outline

1. Continuity Equation for NF Log-Likelihood
2. SDE Basics
3. Diffusion and Score Matching SDEs
4. Probability Flow ODE

# Score Matching SDE

## Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t,$$

$$q(\mathbf{x}_t | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

$$\mathbf{x}_{t-1} = \mathbf{x} + \sigma_{t-1} \cdot \boldsymbol{\epsilon}_{t-1},$$

$$q(\mathbf{x}_{t-1} | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_{t-1}^2 \cdot \mathbf{I})$$

# Score Matching SDE

## Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

$$\mathbf{x}_{t-1} = \mathbf{x} + \sigma_{t-1} \cdot \boldsymbol{\epsilon}_{t-1}, \quad q(\mathbf{x}_{t-1} | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_{t-1}^2 \cdot \mathbf{I})$$

$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot \boldsymbol{\epsilon}, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2) \cdot \mathbf{I})$$

# Score Matching SDE

## Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

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$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot \boldsymbol{\epsilon}, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2) \cdot \mathbf{I})$$

Let's transform this Markov chain into the continuous stochastic process  $\mathbf{x}(t)$  by letting  $T \rightarrow \infty$ :

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \boldsymbol{\epsilon}$$

# Score Matching SDE

## Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

$$\mathbf{x}_{t-1} = \mathbf{x} + \sigma_{t-1} \cdot \boldsymbol{\epsilon}_{t-1}, \quad q(\mathbf{x}_{t-1} | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_{t-1}^2 \cdot \mathbf{I})$$

$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot \boldsymbol{\epsilon}, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2) \cdot \mathbf{I})$$

Let's transform this Markov chain into the continuous stochastic process  $\mathbf{x}(t)$  by letting  $T \rightarrow \infty$ :

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \boldsymbol{\epsilon} \\ &= \mathbf{x}(t - dt) + \sqrt{\frac{\sigma^2(t) - \sigma^2(t - dt)}{dt}} dt \cdot \boldsymbol{\epsilon} \end{aligned}$$

# Score Matching SDE

## Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \epsilon_t, \quad q(\mathbf{x}_t | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

$$\mathbf{x}_{t-1} = \mathbf{x} + \sigma_{t-1} \cdot \epsilon_{t-1}, \quad q(\mathbf{x}_{t-1} | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_{t-1}^2 \cdot \mathbf{I})$$

$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot \epsilon, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2) \cdot \mathbf{I})$$

Let's transform this Markov chain into the continuous stochastic process  $\mathbf{x}(t)$  by letting  $T \rightarrow \infty$ :

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \epsilon \\ &= \mathbf{x}(t - dt) + \sqrt{\frac{\sigma^2(t) - \sigma^2(t - dt)}{dt}} dt \cdot \epsilon \\ &= \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w} \end{aligned}$$

# Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$
$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

## Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$
$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

## Variance Exploding SDE

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$\sigma(t)$  is a monotonically increasing function.

## Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$
$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

## Variance Exploding SDE

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$\sigma(t)$  is a monotonically increasing function.

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

# Diffusion SDE

## Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

# Diffusion SDE

## Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

Let's turn this Markov chain into a continuous stochastic process by letting  $T \rightarrow \infty$  and setting  $\beta_t = \beta(\frac{t}{T}) \cdot \frac{1}{T}$  (where  $dt = \frac{1}{T}$ ):

$$\mathbf{x}(t) = \sqrt{1 - \beta(t)dt} \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon$$

# Diffusion SDE

## Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

Let's turn this Markov chain into a continuous stochastic process by letting  $T \rightarrow \infty$  and setting  $\beta_t = \beta(\frac{t}{T}) \cdot \frac{1}{T}$  (where  $dt = \frac{1}{T}$ ):

$$\begin{aligned} \mathbf{x}(t) &= \sqrt{1 - \beta(t)dt} \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon \approx \\ &\approx \left(1 - \frac{1}{2}\beta(t)dt\right) \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon \end{aligned}$$

# Diffusion SDE

## Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

Let's turn this Markov chain into a continuous stochastic process by letting  $T \rightarrow \infty$  and setting  $\beta_t = \beta(\frac{t}{T}) \cdot \frac{1}{T}$  (where  $dt = \frac{1}{T}$ ):

$$\begin{aligned} \mathbf{x}(t) &= \sqrt{1 - \beta(t)dt} \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon \approx \\ &\approx \left(1 - \frac{1}{2}\beta(t)dt\right) \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon = \\ &= \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w} \end{aligned}$$

# Diffusion SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

# Diffusion SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

## Variance Preserving SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

Variance is preserved as long as  $\mathbf{x}(0)$  has unit variance.

# Diffusion SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

# Diffusion SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

## Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

## Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

We treat the discrete-time methods (NCSN, DDPM) as a special case of the continuous-time methods (VE-SDE, VP-SDE).

# Outline

1. Continuity Equation for NF Log-Likelihood
2. SDE Basics
3. Diffusion and Score Matching SDEs
4. Probability Flow ODE

# Probability Flow ODE

## ODE and Continuity Equation

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$$

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x}, t)}{\partial \mathbf{x}} \right) \Leftrightarrow \frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x}))$$

The only source of randomness is the initial distribution  $p_0(\mathbf{x})$ .

# Probability Flow ODE

## ODE and Continuity Equation

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$$

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The only source of randomness is the initial distribution  $p_0(\mathbf{x})$ .

## SDE and KFP Equation

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

Now there are two sources of randomness: the initial distribution  $p_0(\mathbf{x})$  and the Wiener process  $\mathbf{w}(t)$ .

# Probability Flow ODE

## Theorem

Suppose the SDE  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  induces the probability path  $p_t(\mathbf{x})$ . Then, there exists an ODE with the same probability path  $p_t(\mathbf{x})$ , given by

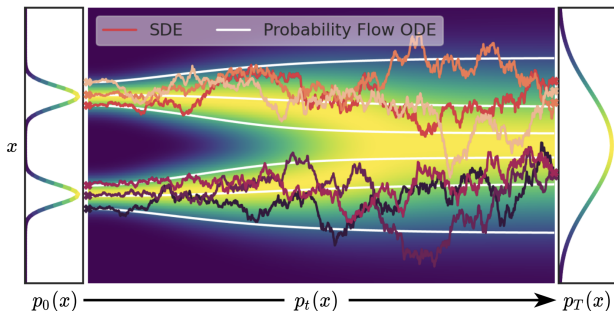
$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

# Probability Flow ODE

## Theorem

Suppose the SDE  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  induces the probability path  $p_t(\mathbf{x})$ . Then, there exists an ODE with the same probability path  $p_t(\mathbf{x})$ , given by

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Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

# Probability Flow ODE

## Theorem

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## Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right)$$

# Probability Flow ODE

## Theorem

Suppose the SDE  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  induces the probability path  $p_t(\mathbf{x})$ . Then, there exists an ODE with the same probability path  $p_t(\mathbf{x})$ , given by

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## Proof

$$\begin{aligned} \frac{\partial p_t(\mathbf{x})}{\partial t} &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = \\ &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} \right] \right) \end{aligned}$$

# Probability Flow ODE

## Theorem

Suppose the SDE  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  induces the probability path  $p_t(\mathbf{x})$ . Then, there exists an ODE with the same probability path  $p_t(\mathbf{x})$ , given by

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## Proof

$$\begin{aligned} \frac{\partial p_t(\mathbf{x})}{\partial t} &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = \\ &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} \right] \right) = \\ &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)p_t(\mathbf{x})\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right] \right) \end{aligned}$$

# Probability Flow ODE

## Theorem

Suppose the SDE  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  induces the probability path  $p_t(\mathbf{x})$ . Then, there exists an ODE with the same probability path  $p_t(\mathbf{x})$ , given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

## Proof

$$\begin{aligned} \frac{\partial p_t(\mathbf{x})}{\partial t} &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = \\ &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} \right] \right) = \\ &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)p_t(\mathbf{x})\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right] \right) = \\ &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) p_t(\mathbf{x}) \right] \right) \end{aligned}$$

# Probability Flow ODE

## Proof (Continued)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) p_t(\mathbf{x}) \right] \right)$$

# Probability Flow ODE

## Proof (Continued)

$$\begin{aligned}\frac{\partial p_t(\mathbf{x})}{\partial t} &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) p_t(\mathbf{x}) \right] \right) = \\ &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})] \right) = -\text{div}(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x}))\end{aligned}$$

# Probability Flow ODE

## Proof (Continued)

$$\begin{aligned}\frac{\partial p_t(\mathbf{x})}{\partial t} &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) p_t(\mathbf{x}) \right] \right) = \\ &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})] \right) = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x}))\end{aligned}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}); \quad \tilde{g}(t) = 0$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt + 0 \cdot d\mathbf{w} = \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}\tilde{g}^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

# Probability Flow ODE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad - \text{SDE}$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \quad - \text{Probability Flow ODE}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$

# Probability Flow ODE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad - \text{SDE}$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \quad - \text{Probability Flow ODE}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$

- $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$  is the continuous-time score function.

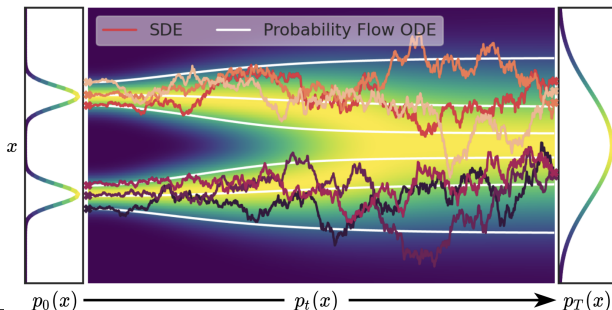
# Probability Flow ODE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad - \text{SDE}$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \quad - \text{Probability Flow ODE}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$

- ▶  $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$  is the continuous-time score function.
- ▶ The ODE produces more stable trajectories.



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

# Why do we need to convert SDEs to PF-ODEs?

- ▶ We could use `SDEsolve` to discretize the SDE.
- ▶ However, this is not efficient (trajectories are not smooth).

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## Efficient Solvers

- ▶ Converting SDEs to PF-ODEs yields more efficient inference.
- ▶ We can apply any `ODEsolve` procedure to reduce the number of inference steps.
- ▶ For example, we can use Heun's / Runge-Kutta methods to solve the PF-ODE instead of Euler's method.
- ▶ In practice, this reduces the number of steps from 100–1000 to 20–50.

# Summary

- ▶ The continuity equation describes the evolution of  $\log p(\mathbf{x}, t)$  over time.
- ▶ An SDE defines a stochastic process with drift and diffusion terms; ODEs are a special case of SDEs.
- ▶ The KFP equation describes the probability dynamics of an SDE.
- ▶ The Langevin SDE preserves a constant probability path.
- ▶ Every SDE admits a corresponding probability flow ODE following the same probability path.